
🔐 What is AWS Certificate Manager (ACM)?

AWS Certificate Manager (ACM) is a fully managed service that lets you **provision, manage, and deploy SSL/TLS certificates** for use with AWS services and your internal connected resources.

These certificates help secure network communications and establish the identity of websites over the internet as well as resources on private networks.

🎯 Key Features

Feature	Description
SSL/TLS Certificate Issuance	Issue public or private certificates (free public certs via ACM).
Automatic Renewal	ACM automatically renews certificates before expiration.
Integrated with AWS Services	Use with ELB, CloudFront, API Gateway, etc., without manual install.
Private CA Support	Manage private CAs and issue internal certificates with ACM Private CA.
No Cost for Public Certs	Public certificates are free; you pay only for services using them.

🧱 Types of Certificates

Type	Description
Public Certificate	Issued by Amazon's public certificate authority. Valid for public domains.

Private Certificate	Issued using **ACM Private CA** for internal domains (e.g., *.corp.internal).

🔧 Common Use Cases

Use Case	Example
Web App on ELB/ALB	Attach ACM certificate to the Load Balancer for HTTPS traffic.
Secure API Gateway	Use SSL/TLS for custom domains on API Gateway.
CloudFront Distribution	Serve HTTPS content via ACM-managed certificates.
IoT and internal services	Use ACM Private CA to manage internal TLS connections.

🛠️ How to Request a Public Certificate (Step-by-Step)

1. **Go to AWS Certificate Manager** in the AWS Console.
2. Click **Request a certificate**.
3. Choose **Request a public certificate**.
4. Enter your domain name(s) (e.g., example.com, *.example.com).
5. Choose **Validation method**:
   - **DNS validation** (preferred)
   - **Email validation**
6. Complete the validation (e.g., add DNS CNAME record if DNS validation).
7. Once validated, the certificate becomes **Issued**.

🧪 Lab Example – Secure a Website with ACM + ALB

1. **Create an ALB** (Application Load Balancer).
2. **Request a public certificate** in ACM for your domain.
3. **Validate the certificate** via DNS.
4. **Attach the ACM certificate** to the HTTPS listener on your ALB.
5. **Point your domain** (e.g., www.example.com) to the ALB using Route 53.

🔄 Certificate Lifecycle in ACM

1. **Request**: Create a certificate request via console, CLI, or SDK.
2. **Validate**: Prove domain ownership (DNS/email).
3. **Issue**: ACM issues the certificate once validated.
4. **Deploy**: Attach to supported AWS services.
5. **Renew**: ACM automatically renews before expiry.
6. **Revalidate** (if needed): DNS/email validation must remain intact for renewal.

🛡️ ACM vs. ACM Private CA

Feature	ACM	ACM Private CA

Use Case	Public-facing apps (HTTPS)	Internal apps, devices, IoT, private services
Cost	Free for certs	Pay per certificate and per private CA
Validation	DNS/Email	No validation required (you control issuance)
Certificate Authority	Amazon's public CA	Your own internal CA

🧠 Best Practices

- Prefer **DNS validation** – it's automated and easier for renewals.
- Use **wildcard certificates** (*.example.com) to cover multiple subdomains.
- Automate certificate creation and deployment using **Terraform, CloudFormation, or Boto3**.
- Monitor certificate expiry with **CloudWatch or AWS Config**.
- Use **ACM Private CA** only when you need full control over the PKI lifecycle.

📚 Useful CLI Commands

```
# Request a certificate
aws acm request-certificate \
  --domain-name example.com \
  --validation-method DNS

# List certificates
aws acm list-certificates
```

```
# Describe a certificate
aws acm describe-certificate --certificate-arn <ARN>

# Delete a certificate
aws acm delete-certificate --certificate-arn <ARN>
```

📝 Summary Notes

Topic	Details
Service	AWS Certificate Manager (ACM)
Purpose	Issue and manage SSL/TLS certificates
Public Certs	Free, for use with ELB, CloudFront, API Gateway
Private Certs	Via ACM Private CA (paid service)
Auto-Renewal	Yes, for both public and private certs
Integration	ELB, CloudFront, API Gateway, etc.
Validation Methods	DNS (preferred), Email
